9.1 Purpose

The purpose of this task is to implement the complete electricity billing program and test it with normal inputs, boundary cases, meter rollover, and invalid inputs.

9.2 Testing

The program should be tested with zero units, exactly 100 units, exactly 200 units, and consumption above 200 units. It should also test meter rollover from 65535 to 0 and reject invalid readings.

9.3 Alternative Concept and Complexity Analysis

Using fixed "if-else" statements is simple when there are only a few tariff slabs. Using arrays and a loop makes the program easier to extend when more slabs are added. For (s) tariff slabs, the loop takes (O(s)) time and uses (O(s)) space for the tariff arrays.

Conclusion

Testing verifies that the billing program calculates charges correctly and handles boundary conditions and invalid inputs safely.

```c
#include <stdio.h>

#define MAX_READING 65535U
#define METER_MODULUS 65536U
#define FIXED_CHARGE_PAISE 5000LL
#define RATE2_PAISE 225LL
#define RATE3_PAISE 450LL

long long compute_bill(unsigned int units)
{
    long long bill = FIXED_CHARGE_PAISE;

    if (units > 100 && units <= 200)
        bill += (long long)(units - 100) * RATE2_PAISE;
    else if (units > 200) {
        bill += 100LL * RATE2_PAISE;
        bill += (long long)(units - 200) * RATE3_PAISE;
    }

    return bill;
}

int main(void)
{
    unsigned int previous, current, units;
    unsigned int tests[] = {0, 100, 150, 200, 350};

    printf("BILLING TEST TABLE\n");
```

```
    for (int i = 0; i < 5; i++) {
        long long bill = compute_bill(tests[i]);
        printf("%u units -> Rs %lld.%02lld\n",
               tests[i], bill / 100, bill % 100);
    }

    printf("\nEnter previous and current meter readings: ");

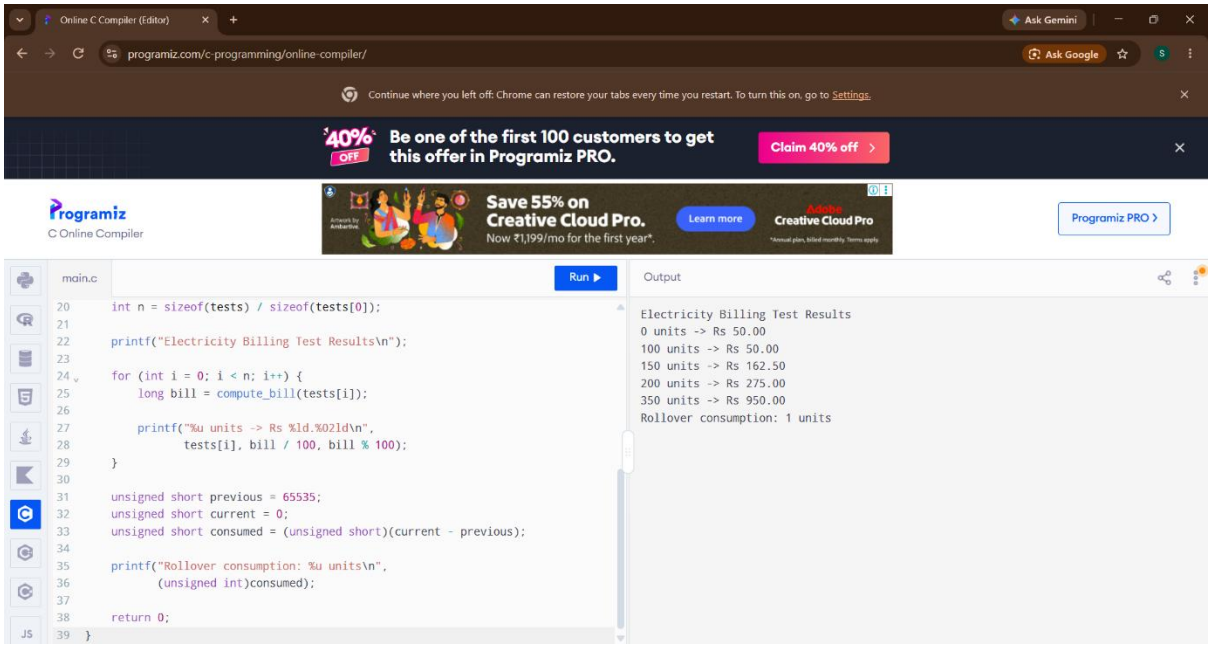
    if (scanf("%u %u", &previous, &current) != 2 ||
        previous > MAX_READING || current > MAX_READING) {
        printf("Invalid meter reading.\n");
        return 1;
    }

    if (current >= previous)
        units = current - previous;
    else
        units = METER_MODULUS - previous + current;

    long long bill = compute_bill(units);

    printf("Units consumed: %u\n", units);
    printf("Total bill: Rs %lld.%02lld\n",
           bill / 100, bill % 100);

    return 0;
}
```


Online C Compiler (Editor)
Ask Gemini
programiz.com/c-programming/online-compiler/
Ask Google
Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to Settings.
40% OFF
Be one of the first 100 customers to get this offer in Programiz PRO.
Claim 40% off
Programiz
C Online Compiler
Save 55% on Creative Cloud Pro.
Now ₹1,199/mo for the first year*.
Learn more
Adobe Creative Cloud Pro
Programiz PRO
main.c
Run
Output
int n = sizeof(tests) / sizeof(tests[0]);
printf("Electricity Billing Test Results\n");
for (int i = 0; i < n; i++) {
long bill = compute_bill(tests[i]);
printf("%u units -> Rs %ld.%02ld\n",
tests[i], bill / 100, bill % 100);
}
unsigned short previous = 65535;
unsigned short current = 0;
unsigned short consumed = (unsigned short)(current - previous);
printf("Rollover consumption: %u units\n",
(unsigned int)consumed);
return 0;
}
Electricity Billing Test Results
0 units -> Rs 50.00
100 units -> Rs 50.00
150 units -> Rs 162.50
200 units -> Rs 275.00
350 units -> Rs 950.00
Rollover consumption: 1 units